Solving compound inequalities

1

a $3 < 6x + 7$ and $6x + 7 < 9$ **b** And

2

a $x < 100$ **b** $100 \leq x \leq 129$ **c** $x \geq 190$

3

a No **b** No **c** Yes **d** No

4

a No **b** Yes

5 4

Step Number	Step
4	Divide each part by 6.
3	Simplify by collecting constant terms.
2	Subtract 8 from all three parts.
5	Simplify any divisions.
1	Write the given compound inequality.

6 $23 \leq F \leq 95$

7 $0.5 < t < 1.5$

8 $45 \leq x < 95$

9 $5 \leq t \leq 8$

10 $84 \leq x \leq 100$

11

a Yes **b** No

12

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13

a

i $x < -9$

ii $x > 6$

iii $x < -9$ or $x > 6$

iv



b

i $x > -4$

ii $x > -2$

iii $x > -4$

iv



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c

i $x \leq 2$

ii $x \leq 10$

iii $x \leq 10$

iv



d

i $x < 11$

ii $x > 1$

iii All real x.

iv



e

i $x > 1$

ii $x > 5$

iii $x > 1$

iv



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f

i $x > 1$

ii $x > 5$

iii $x > 1$

iv



Additional questions

14

a

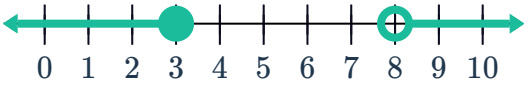


b



15

a



b



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16

a

i $x > -5$ and $x < 8$.

ii



iii $(-5, 8)$

b

i $x > -7$ and $x \leq 5$

ii



iii $(-7, 5]$

17

a There are no possible values for x.

b $-5 \leq x < 3$

18

a $x < -3$ or $x > 6$

b $-\infty < x < \infty$

19 Step 2. Carlton divided the compound inequality by -1 to make all of the signs positive, but didn't correctly switch the inequality symbols. This line should be $6 < 3x \leq 18$.

20 Skyler correctly solved both parts of the compound inequality to get $x \leq 3$ and $x > 9$, however he plotted the inequality using **or** instead of **and**. The correct answer has no points on the numberline.

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